

$$Q1) \quad \hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$$

$$2) \text{ Consider } H = H_0 - \alpha \hat{x}.$$

2nd order PT to estimate GSE.

Compare w/ actual GSE (CTS).

$$E = E_0^0 - \alpha E_0^1 + \alpha^2 E_0^2$$

$$E_0^1 = -\alpha \langle n^0 | x | n^0 \rangle$$

$$E_0^2 = \alpha^2 \sum_{n \neq m} \frac{|\langle n^0 | x | m^0 \rangle|^2}{E_n^0 - E_m^0}$$

H_0 has ev's $\hbar\omega(n + \frac{1}{2})$ or energies, states determined by raising/lowering ops:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (a^\dagger + a)$$

When acting on $n=0$, $a|0\rangle = 0$; $\langle 0|a^\dagger = 0$

so first order correction vanishes.

$$E_n^2 = \gamma^2 \sum_{n \neq m} \frac{|\langle n | \cdot \frac{\hbar}{2m\omega} G^{(2)} | m \rangle|^2}{E_n^2 - E_m^2}$$

$$E_n^0 - E_m^0 = \hbar\omega (n-m)$$

$$= \gamma^2 \frac{\hbar^2}{4m^2\omega^2} \cdot \frac{1}{\hbar\omega} \sum_{n \neq m} \frac{|\langle n | G^{(2)} | G^{(2)} | m \rangle|^2}{n-m}$$

= ...

$$= \frac{\gamma^2 \hbar}{4m^2\omega^3} \left[\sum_{n \neq m} \frac{(m+1)(m+2) \delta_{n,m+2}}{n-m} + \frac{m(m-1) \delta_{n,m-2}}{n-m} \right]$$

$m=n-2$ $m=n+2$

$$\frac{(n-1)n}{n-(n-2)} + \frac{(n+2)(n+1)}{n-(n+2)}$$

$$= \frac{n^2-n}{2} + \frac{n^2+3n+2}{-2}$$

$$= \frac{n^2 - n^2 - n - 3n - 2}{2}$$

$$= -(2n+1)$$

At GSE, $E_0^2 = - \frac{\gamma^2 \hbar}{4m^2\omega^3}$

Hence

$$E_0 = \frac{\hbar\omega}{2} + \frac{\gamma}{\hbar m \omega} - \frac{\gamma^2 \hbar}{4m^2\omega^3}$$

$$Q4) \quad H = H_0 + W$$

$$H_0 |n\rangle = E_n^0 |n\rangle$$

$$E_2^0 = E_1^0 \quad \text{and} \quad \langle n | W | m \rangle = W^{nm}$$

First order E correction: diagonalize W^{nn}

$$\begin{pmatrix} \epsilon & \epsilon \\ \epsilon & \epsilon \end{pmatrix} = \epsilon \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \rightarrow \text{char}(W) = \lambda(\lambda - 2\epsilon) = 0$$

$$2) \text{ so } \lambda = 0: \quad E = E_1^0$$

$$\lambda = 2\epsilon: \quad E = E_1^0 + 2\epsilon$$

$$b) \text{ EV's are } \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ or } \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \text{ Normalize}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ or } \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \rightarrow \begin{matrix} \nearrow \equiv |1'\rangle \\ \searrow \equiv |2'\rangle \end{matrix}$$

but this is only at 0th order.

First order w/ correction)

Obviously 0th order correction must diagonalize

$\hat{H}_1$ ($|n, k_n\rangle$) in eigenstates.

$$\begin{aligned} \text{We have } (H_0 + \lambda H_1) (|n, k_n\rangle^0 + \lambda |n, k_n\rangle^1 + \dots) \\ = (E_n^0 + \lambda E_n^1 + \dots) (|n, k_n\rangle^0 + \lambda |n, k_n\rangle^1 + \dots) \end{aligned}$$

At $\mathcal{O}(\lambda)$:

$$\begin{aligned} H_1 |n, k_n\rangle^0 + H_0 |n, k_n\rangle^1 &= E_n^0 |n, k_n\rangle^1 + E_n^1 |n, k_n\rangle^0 \\ \Rightarrow \langle m, k_m | n, k_n \rangle^0 E_n^1 &= (H_0 - E_n^0) |n, k_n\rangle^1 \\ &\quad + \langle m, k_m | H_1 | n, k_n \rangle^0 \end{aligned}$$

Cases: $n \neq m$ (no degeneracy)

$$\sum_m \cancel{\delta_{m,n}} E_n^1 = \langle m, k_m | n, k_n \rangle^1 (E_m^0 - E_n^0) + \langle m, k_m | H_1 | n, k_n \rangle^0$$

$$\text{or } \langle m, k_m | n, k_n \rangle^1 = \frac{\langle m, k_m | H_1 | n, k_n \rangle^0}{E_n^0 - E_m^0} \quad [n \neq m]$$

↑ includes
non-degenerate
region

$n = m, k_n \neq k_m$ (degeneracy)

$$\sum_m \cancel{\delta_{m,n}} \cancel{\delta_{k_m, k_n}} E_n^1 = \langle m, k_m | n, k_n \rangle^1 (E_{m, k_m}^0 - E_{n, k_n}^0) + \langle m, k_m | H_1 | n, k_n \rangle^0$$

$$\text{or } \langle m, k_m | n, k_n \rangle^1 = \frac{\langle m, k_m | H_1 | n, k_n \rangle^0}{E_{n, k_n}^0 - E_{m, k_m}^0}$$

↑
degenerate
region only

Under completeness

$$|n, k_n\rangle = \sum_{m, k_m, \dots} |m, k_m\rangle^0 \langle m, k_m | n, k_n \rangle^1$$

$$= \sum_{m \neq n} \frac{\langle m, k_m | \hat{H}_1 | n, k_n \rangle}{E_n^0 - E_m^0} |m, k_m\rangle^0$$

↑
nondegenerate

$$+ \sum_{\substack{m=n, \\ k_m \neq k_n}} \frac{\langle m, k_m | H_1 | n, k_n \rangle}{E_{n, k_n}^{(1)} - E_{n, k_m}^{(1)}} |m, k_m\rangle^0$$

↑
degenerate

Back to problem:

$$|1'\rangle = \frac{1}{\sqrt{2}} (|1\rangle - |2\rangle)$$

are 0th order corrections

$$|2'\rangle = \frac{1}{\sqrt{2}} (|1\rangle + |2\rangle)$$

Degenerate term has $\frac{0 \cdot |1'\rangle}{E_n^0 - E_n^0} \rightarrow \infty$

and

$$\frac{2E |2'\rangle}{E_n^0 - E_n^0} \rightarrow \infty$$

$$\text{So } \sum_K \frac{\langle K | W | 1' \rangle}{E_1^0 - E_K^0} = \sum_{K \neq 1, 2} \frac{\frac{1}{\sqrt{2}} (W^{K1} - W^{K2})}{E_1^0 - E_K^0}$$

... .. Sim for 2nd term

Hence

$$|\psi_2\rangle = |1'\rangle + \sum_{K \neq 1, 2} \frac{\frac{1}{\sqrt{2}} (W^{K1} - W^{K2})}{E_1^0 - E_K^0} |K\rangle \quad \left(\begin{array}{l} 1^{st} \text{ order} \\ \text{w/f corrections} \end{array} \right)$$

$$|\psi_2'\rangle = |2'\rangle + \sum_{K \neq 1, 2} \frac{\frac{1}{\sqrt{2}} (W^{K1} + W^{K2})}{E_2^0 - E_K^0} |K\rangle$$

c) Energy E vs to 2nd order

Ans

$$E_1 = E_1^0 + 0 + \sum_{k \neq 1,2} \frac{|(W^{k1} - W^{k2})|^2}{E_1^0 - E_k^0}$$

$$E_2 = E_2^0 + 2\varepsilon + \sum_{k \neq 1,2} \frac{|(W^{k1} + W^{k2})|^2}{E_2^0 - E_k^0}$$

Time-Dependent (Q5):

$$V(x,t) = D x e^{-x^2/\kappa^2} \cos(\omega_0 t) + \frac{1}{2} m \omega_0^2 x^2.$$

At $t = -\infty$ particle is in its ground state. For $D \ll 1$,

find probability at $t = \infty$ that a measurement would yield first excited state?

$$\hat{H} = \hat{H}_0 + D \hat{H}_1$$

$$= \underbrace{\frac{\hat{p}_x^2}{2m} + \frac{1}{2} m \omega_0^2 \hat{x}^2}_{\hat{H}_0} + \underbrace{D \hat{x} e^{-x^2/\kappa^2} \cos \omega_0 t}_{D \hat{H}_1(t)}.$$

The $\hat{H}_0$ is SHO, with eigenstates $|n\rangle$; $E_n^0 = \hbar \omega_0 (n + \frac{1}{2})$.

$$\text{and } \hat{x} = \sqrt{\frac{\hbar}{2m\omega_0}} (\hat{a} + \hat{a}^\dagger).$$

$$\frac{E_1 - E_0}{\hbar} = \frac{\hbar\omega}{\hbar} \left(\frac{3}{2} - \frac{1}{2} \right) = \omega,$$

$$\text{We have that } P_{i \rightarrow f}(t, t_0) = |\langle 1 | 0 \rangle|^2.$$

$$\begin{aligned} P_{i \rightarrow f}(t_0, t) &= \frac{D^2}{\hbar^2} \left| \int_{t_0}^t e^{i \frac{E_1 - E_0}{\hbar} t'} \left(\frac{1}{\sqrt{2m\omega_0}} \right) e^{-t'^2/\tau^2} \cos(\omega_d t') dt' \right|^2 \\ &= \frac{D^2}{\hbar^2} \cdot \frac{\hbar}{2m\omega_0} \left| \int_{t_0}^t e^{i\omega_0 t'} e^{-t'^2/\tau^2} \cos(\omega_d t') dt' \right|^2 \\ &= \frac{D^2}{2m\omega_0 \hbar} \frac{1}{4} \left| \int_{-\infty}^{\infty} dt' e^{i\omega_0 t' - t'^2/\tau^2} (e^{i\omega_d t'} + e^{-i\omega_d t'}) \right|^2 \\ &= \frac{D^2}{2m\omega_0 \hbar} \frac{1}{4} \left| \int_{-\infty}^{\infty} dt' \left[e^{i(\omega_0 + \omega_d)t' - t'^2/\tau^2} + e^{i(\omega_0 - \omega_d)t' - t'^2/\tau^2} \right] \right|^2 \end{aligned}$$

Let's do the integral

$$\int_{-\infty}^{\infty} dt' e^{i\omega_{\pm} t' - t'^2/\tau^2} : \text{complete the square:}$$

$$-\frac{t'^2}{\tau^2} + i\omega_{\pm} t' = -\frac{1}{\tau^2} \left(t'^2 - i\omega_{\pm} \tau^2 t' \right)$$

$$= -\frac{1}{\tau^2} \left(t'^2 - i\omega_{\pm} \tau^2 t' + \left(\frac{i\omega_{\pm} \tau^2}{2} \right)^2 - \left(\frac{i\omega_{\pm} \tau^2}{2} \right)^2 \right)$$

$$= -\frac{1}{\tau^2} \left(t' - \frac{i\omega_{\pm} \tau^2}{2} \right)^2 + \frac{1}{\tau^2} \left(\frac{i\omega_{\pm} \tau^2}{2} \right)^2$$

$$= -\frac{1}{\tau^2} \left(t' - \frac{i\omega_{\pm} \tau^2}{2} \right)^2 - \frac{\omega_{\pm}^2 \tau^2}{4}$$

$$\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}}$$

$$= e^{-\frac{\omega_+^2 \tau^2}{4}} \int_{-\infty}^{\infty} dt e^{-\frac{1}{\tau^2} \left(1 - \frac{i\omega_{\pm} \tau^2}{2}\right)^2}$$

$$u = t - \frac{i\omega_{\pm} \tau^2}{2}$$

$$du = dt$$

$$= e^{-\omega_+^2 \tau^2 / 4} \int_{-\infty}^{\infty} du e^{-\frac{u^2}{\tau^2}}$$

$$= e^{-\omega_+^2 \tau^2 / 4} \left[\sqrt{\pi \tau^2} \right]$$

$$= \tau \sqrt{\pi} e^{-\omega_+^2 \tau^2 / 4}$$

Hence

$$P_{i \rightarrow f}(+ \rightarrow +) = \frac{D^2}{8m\omega_0 \hbar} \left| \tau \sqrt{\pi} e^{-\omega_+^2 \tau^2 / 4} + \tau \sqrt{\pi} e^{-\omega_-^2 \tau^2 / 4} \right|^2$$

$$= \frac{D^2 \tau^2 \pi}{8m\omega_0 \hbar} \left| e^{-\omega_+^2 \tau^2 / 4} + e^{-\omega_-^2 \tau^2 / 4} \right|^2$$

$$= \frac{D^2 \tau^2 \pi}{8m\omega_0 \hbar} \left| e^{-(\omega_0 + \omega_d)^2 \tau^2 / 4} + e^{-(\omega_0 - \omega_d)^2 \tau^2 / 4} \right|^2$$

This is a maximum when

$$\omega_0 = \pm \omega_d$$

and goes into

$$P_{\text{tot}} \approx \frac{D^2 \tau^2 \pi}{8m\omega_0 \hbar}$$

